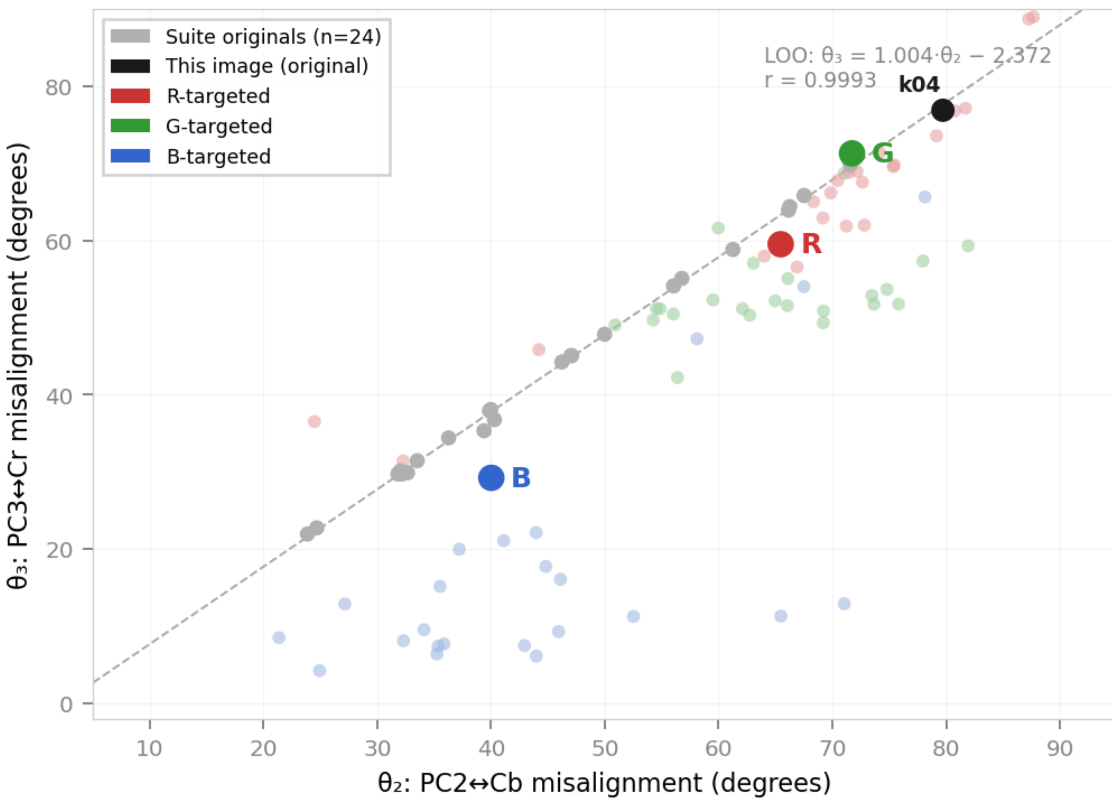
# Compression Profile: KODIM04 512x768 | 24-bit RGB | PNG (lossless)

portrait of a girl in red | Weakly One-Dimensional (PC1 = 86.87%)

## 1. Geometric Displacement Under Directional Perturbation




## 2. Pairwise Correlation and Angular Misalignment

| Version | R-G | R-B | G-B | Avg \|r\| | theta2 | theta3 | LOO dev |
|---|---|---|---|---|---|---|---|
| Original | 0.6004 | 0.6850 | 0.9568 | 0.747 | 79.72° | 76.92° | -0.75° |
| R-targeted | 0.1604 | 0.0608 | 0.9726 | 0.398 | 65.46° | 59.54° | -3.81° |
| G-targeted | -0.2929 | 0.9460 | -0.1132 | 0.451 | 71.73° | 71.33° | +1.69° |
| B-targeted | 0.9039 | 0.1804 | -0.1765 | 0.420 | 40.02° | 29.23° | -8.58° |

R-targeted isolates red (R-G, R-B < 0.16) while preserving G-B coupling at 0.973.
G-targeted isolates green (R-G, G-B < 0.29) while preserving R-B at 0.946.
B-targeted preserves R-G at 0.904 while driving R-B and G-B near zero.
All three perturbation directions depart from the LOO constraint manifold (r = 0.9993).

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## 3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

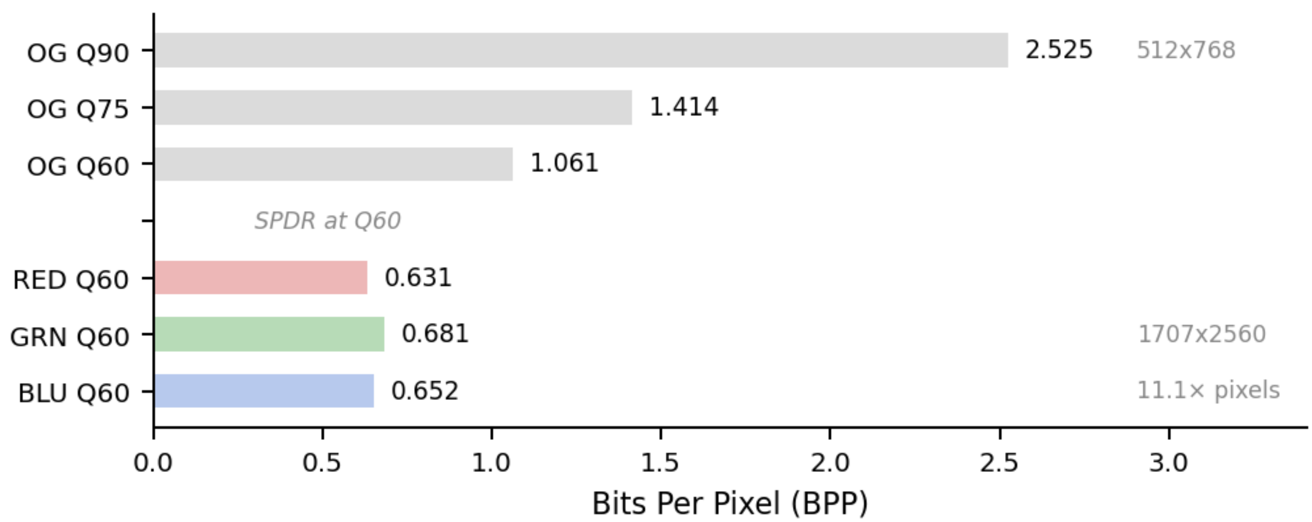
**ORIGINAL (kodim04)**

| Version | Format | Resolution | Size | BPP | Pixels |
|---|---|---|---|---|---|
| PNG (lossless) | PNG | 512x768 | 622.5 KB | 12.969 | 393,216 |
| JPEG Q90 | JPEG | 512x768 | 121.2 KB | 2.525 | 393,216 |
| JPEG Q75 | JPEG | 512x768 | 67.9 KB | 1.414 | 393,216 |
| JPEG Q60 | JPEG | 512x768 | 50.9 KB | 1.061 | 393,216 |

**REDISTRIBUTED (kodim04 SPDR at Q60)**

| Version | Format | Resolution | Size | BPP | Pixels |
|---|---|---|---|---|---|
| RED Q60 | JPEG | 1707x2560 | 336.8 KB | 0.631 | 4,369,920 |
| GRN Q60 | JPEG | 1707x2560 | 363.1 KB | 0.681 | 4,369,920 |
| BLU Q60 | JPEG | 1707x2560 | 347.6 KB | 0.652 | 4,369,920 |

## 4. Quality Ladder: SPDR Q60 vs Original Q90




All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

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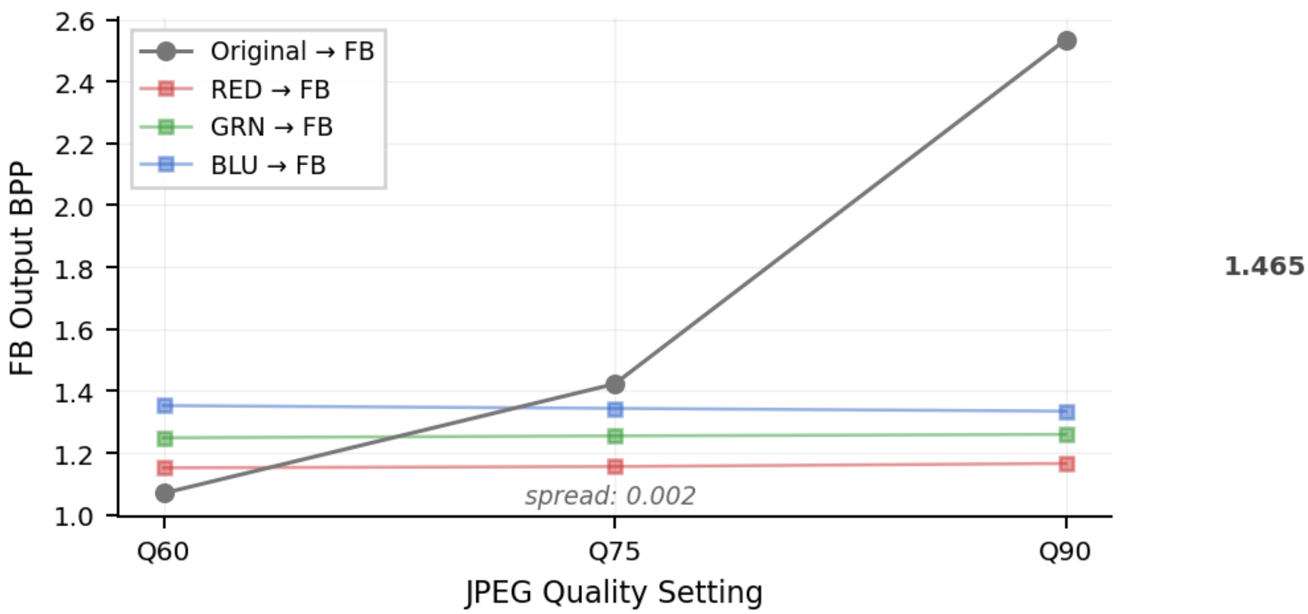
## 5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

| Pipeline | Input Res | Output Res | FB BPP | BPP reduction |
|---|---|---|---|---|
| OG PNG → FB | 512x768 | 512x768 | 2.312 | — |
| R-targeted → FB | 1707x2560 | 1366x2048 | 1.166 | 49.6% |
| G-targeted → FB | 1707x2560 | 1366x2048 | 1.260 | 45.5% |
| B-targeted → FB | 1707x2560 | 1366x2048 | 1.335 | 42.3% |
| ABC (no perturbation) → FB | 1707x2560 | 1366x2048 | 1.257 | 45.6% |

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.7466 vs 0.747 original). Comparable BPP reduction was not observed in the ABC control.

## 6. Quality Parameter Sensitivity Through Facebook




Original: Q60 → Q90 produces a 1.465 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.002 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60–90 BPP variation through the measured pipeline.

## 7. Pipeline Summary — kodim04

| | | |
|---|---|---|
| Best BPP (Q60, direct) | **0.631** | 1707x2560, 4,369,920 px |
| Best BPP through FB | **1.166** | 1366x2048 output |
| vs OG Q90 → FB | **54.0%** | SPDR Q60→FB vs OG Q90→FB |
| Pixel ratio | **11.1×** | 4,369,920 vs 393,216 |